

Density Functional Theory: Treatment of the Electronic Structure

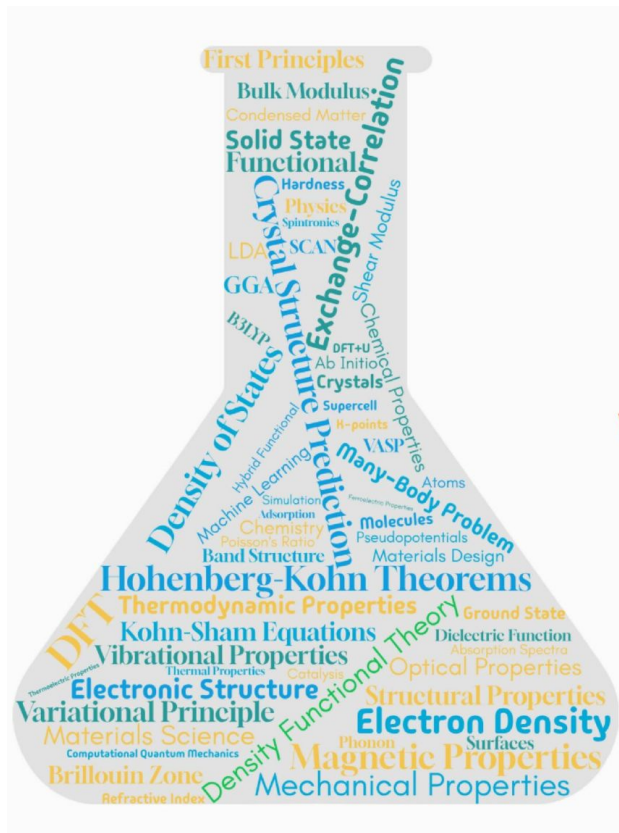
PWF Summer school on Theoretical and Computational Biophysics

University of Colombo

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Dr. Marta Monti

What is DFT?



Density functional theory (DFT) is a quantum-mechanical method used to describe the (ground-state) **electronic structure of many-body systems** as atoms, molecules, or solids.

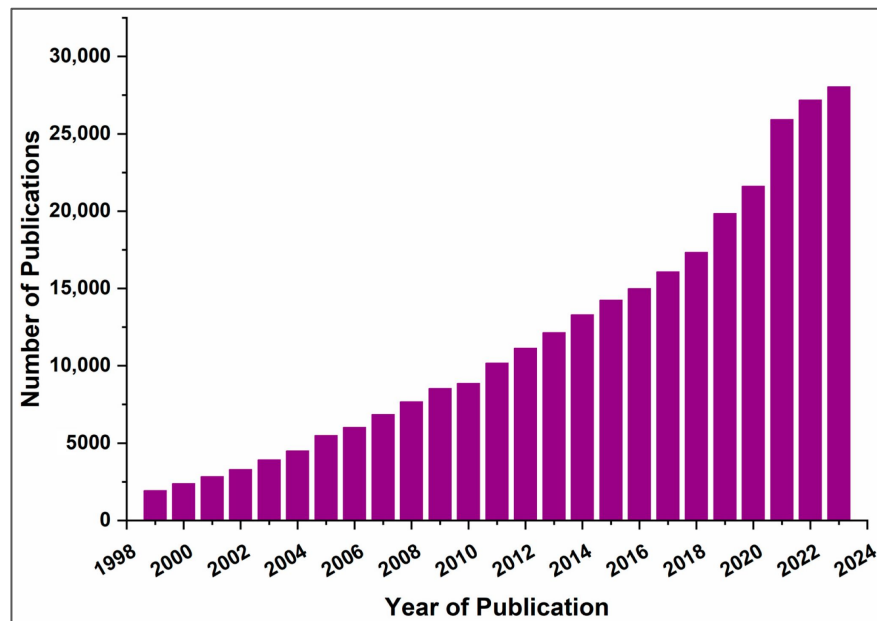
- The central quantity in DFT is the **electron density** $\rho(r)$
- From the electronic structure we can determine the **energy of the system** and many other properties.

What can we learn from DFT?

DFT is widely used in chemistry, biophysics, materials science, and machine learning.

A DFT calculation can provide us with:

- Optimized molecular structures
- Total energies
- Forces on the nuclei
- Reaction energies and activation barriers
- Electron density and charge distribution
- Electronic structure (orbitals, energy levels)
- Vibrational properties and spectra



The quantum-mechanical problem

In principle, by solving the **Schrödinger equation** we can obtain the **energy** and **wavefunction** of a quantum system:

$$\hat{H}\Psi = E\Psi$$

Hamiltonian operator, containing
the different energy contributions

Wavefunction of the system

Total energy

For an N -electron system, however, solving the Schrödinger equation exactly becomes **extremely difficult**

The quantum-mechanical problem

The N -electron system is described by the wavefunction:

$$\Psi(\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_N) \quad \mathbf{X}_i = (\underbrace{x_i, y_i, z_i}_{\substack{\swarrow \\ \text{Cartesian coordinates}}}, \underbrace{\omega_i}_{\substack{\downarrow \\ \text{Spin coord.}}})$$

The many-electron wavefunction depends on the spatial and spin coordinates of N electrons



Problem with $3N$ spatial variables + N spin variables

From the wavefunction to the electron density

In DFT, the N-electron system is described by the electron density:

$$\rho(\mathbf{r}) \quad \mathbf{r} = \underline{(x, y, z)}$$


which depends only on the **three** spatial coordinates

From $3N$ spatial
variables



To 3 spatial
variables

What is the electron density?

The electron density describes the probability of finding electrons in an **infinitesimal region of space** around a given point $\mathbf{r}$:

$$\rho(\mathbf{r})dV$$

$$\rho(\mathbf{r}) = \underbrace{N}_{\text{electrons are indistinguishable}} \sum_{\omega_1} \int |\Psi(\mathbf{r}, \omega_1, \mathbf{X}_2, \dots, \mathbf{X}_N)|^2 d\mathbf{X}_2 \dots d\mathbf{X}_N$$

electrons are **indistinguishable**:
 N accounts for the possibility of finding any electron at $\mathbf{r}$

integration over the spatial coordinates and summation over the spin coordinates of the remaining $N-1$ electrons

By integrating the density over all space, we obtain the total number of electrons:

$$\int \rho(\mathbf{r})d\mathbf{r} = N$$

What is a functional?

Fundamental object of DFT → **functional**, F .

A functional is a mathematical object that takes a function as input and returns a scalar:

$$f: \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x) \quad \text{function}$$

$$F: \mathcal{F} \rightarrow \mathbb{R}, f \mapsto F[f] \quad \text{functional}$$

In DFT, the relevant **function** is the electron density and the total energy is the **functional**:

$$\rho(\mathbf{r})$$

$$E[\rho]$$

Hohenberg-Kohn theorems

A bit of historical context:

- DFT is based on the two theorems proposed by Hohenberg and Kohn (HK) in 1964 (Phys. Rev., 136, 1964)
 - Some of the limitations of the HK theorems were overcome by the constrained-search formulation introduced by Levy and Lieb (PNAS, 76, 1979; Int. J. Quantum. Chem., 24, 1983)
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First HK theorem → it proves that there is a one-to-one correspondence between the ground-state (GS) density and the external potential:

$$\rho_{GS} \leftrightarrow V_{ext}$$

Second HK theorem → it establishes that the GS energy can be written as a universal functional of the density, obeying a variational principle:

$$\begin{aligned} E[\rho] &\geq E_{GS} \\ E[\rho_{GS}] &= E_{GS} \end{aligned}$$

The energy functional

The energy functional contains two main quantities:

$$E[\rho] = F[\rho] + \int V_{ext}(\mathbf{r})\rho(\mathbf{r})d\mathbf{r}$$


In principle: V_{ext} can include any external potential acting on the electrons

In practice: the nuclear attraction dominates and V_{ext} takes the following form:

$$V_{ext}(\mathbf{r}) = \sum_K^{Nucl} -\frac{Z_K}{|\mathbf{r} - \mathbf{R}_K|} \longrightarrow \text{Coulomb attraction}$$

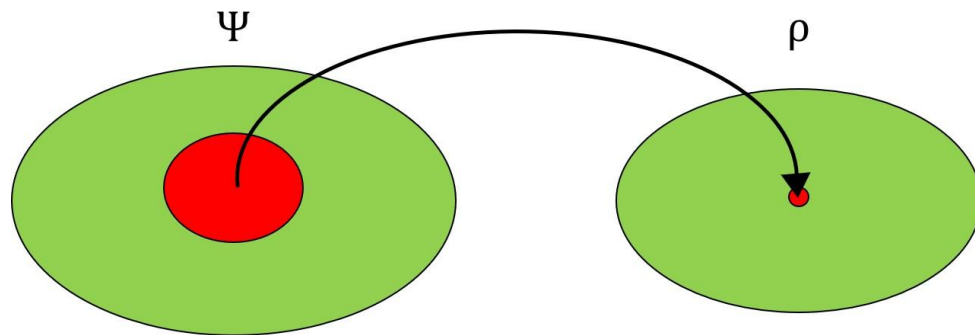
Once we have ρ , V_{ext} is straightforward to compute (we know Z_K and $\mathbf{R}_K$)

The energy functional

Within the Levy-Lieb formalism, the functional $F[\rho]$ is defined as:

$$F[\rho] = \min_{\Psi \rightarrow \rho} \langle \Psi | \hat{T} + \hat{V}_{ee} | \Psi \rangle$$

- Ψ : N-electron antisymmetric wavefunction.
- Multiple Ψ that produce the same ρ .
- Levy tells us to pick Ψ that minimizes the kinetic and electron-electron energy.



$$F[\rho] = \langle \Psi_{min} | \hat{T} | \Psi_{min} \rangle + \langle \Psi_{min} | \hat{V}_{ee} | \Psi_{min} \rangle$$

The energy functional

Minimizing $E[\rho]$ with respect to ρ provides the density and the energy of the GS:

$$E[\rho] = \langle \Psi_{min} | \hat{T} | \Psi_{min} \rangle + \langle \Psi_{min} | \hat{V}_{ee} | \Psi_{min} \rangle + \int V_{ext}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r}$$

Problem...

The energy functional

Minimizing $E[\rho]$ with respect to ρ provides the density and the energy of the GS:

$$E[\rho] = \underbrace{\langle \Psi_{min} | \hat{T} | \Psi_{min} \rangle + \langle \Psi_{min} | \hat{V}_{ee} | \Psi_{min} \rangle} + \int V_{ext}(\mathbf{r})\rho(\mathbf{r})d\mathbf{r}$$

We can't compute it

- We should solve a minimization over N-electron wavefunctions.
- We have a formal justification of the theory and a rigorous definition of the objects needed but we need an approach to approximate $E[\rho]$.
- Direct approach: **Thomas-Fermi** → functionals are approximated (electron gas) and the density is varied in order to reach the minimum of $E[\rho]$. It is considered the precursor of modern DFT; too crude for modern applications.
- Indirect approach: **Kohn-Sham** → our practical DFT approach.

The Kohn-Sham approach

We consider a fictitious system of non-interacting electrons ($V_{ee} = 0$) whose electron density is equal to that of the real system (interacting electrons).

Impractical problem:
treating a many-body system of
interacting e^- within V_{ext}



Practical solution:
treating a system of non-interacting
 e^- within an effective potential (V_{KS})

We can write the KS Hamiltonian as a sum of mono-electronic Hamiltonians:

$$H_{KS} = \sum_{i=1}^N \left(-\frac{1}{2} \nabla_i^2 \right) + \sum_{i=1}^N V_{KS}(r_i) = \sum_{i=1}^N h_{KS}(i)$$


V_{KS} is defined so that ρ fictitious system = ρ real system

The Kohn-Sham approach

The GS wavefunction of the KS system can be constructed with a single Slater determinant:

$$\Psi_{KS} = \frac{1}{\sqrt{N!}} \det | \chi_1 \chi_2 \dots \chi_N |$$

χ_i = spin-orbital satisfying the time-independent Schrödinger equation: $\hat{h}_{KS} \chi_i = \varepsilon_i \chi_i$

The KS density can be calculated from the spatial part of the spin-orbitals:

$$\rho(\mathbf{r}) = 2 \sum_{i=1}^{N/2} \varphi_i^*(\mathbf{r}) \varphi_i(\mathbf{r})$$

To ensure the previous condition for V_{KS} , we minimize the energy functional imposing this constraint on the density:

$$\int \rho(\mathbf{r}) d\mathbf{r} = N$$

The density integrates the correct number of electrons

The Kohn-Sham approach

The constrained minimization of $E[\rho]$ is performed using the Lagrangian multipliers method:

$$\frac{\delta}{\delta \rho} \left(E[\rho] - \lambda \int \rho(\mathbf{r}) d\mathbf{r} \right) = 0$$

λ enforces the electron number while we vary ρ

For the interacting system, we rewrite $E[\rho]$ as:

$$E[\rho] = \underbrace{T_0[\rho]}_{\text{Non-interacting kinetic energy functional}} + \underbrace{J[\rho]}_{\text{Classical electrostatic interaction}} + \underbrace{E_{XC}[\rho]}_{\text{Exchange and correlation functional}} + \int V_{ext}(\mathbf{r}) \rho(\mathbf{r}) d\mathbf{r}$$

Non-interacting kinetic energy functional

$$T_0[\rho] = \sum_{i=1}^N \left\langle \chi_i \left| -\frac{1}{2} \nabla^2 \right| \chi_i \right\rangle$$

Classical electrostatic interaction

$$J[\rho] = \frac{1}{2} \int \int \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'$$

Exchange and correlation functional

$$E_{XC}[\rho] = T[\rho] - T_0[\rho] + V_{ee}[\rho] - J[\rho]$$

More in detail in a few slides!

The Kohn-Sham approach

The constrained minimization of $E[\rho]$ is performed using the Lagrangian multipliers method:

$$\frac{\delta}{\delta\rho} \left(E[\rho] - \lambda \int \rho(\mathbf{r}) d\mathbf{r} \right) = 0$$

For the interacting (real) system:

$$\frac{\delta T_0}{\delta\rho} + \underbrace{\int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{xc}}{\delta\rho} + V_{ext}(\mathbf{r})}_{\text{We imposed } \rho_{\text{real}} = \rho_{\text{KS}}} = \lambda$$

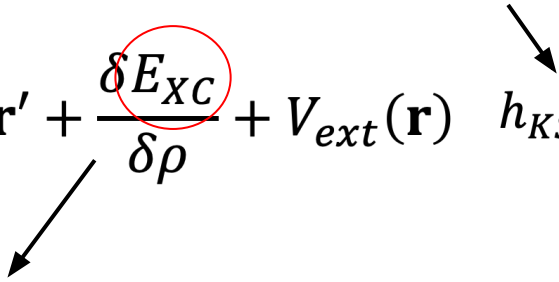
For the fictitious (KS) system:

$$\frac{\delta T_0}{\delta\rho} + \underbrace{V_{KS}(\mathbf{r})}_{\text{We imposed } \rho_{\text{real}} = \rho_{\text{KS}}} = \lambda$$

Two stationary equations for that same density

The Kohn-Sham approach

We finally have an expression for V_{KS} and we can solve the mono-electronic KS equations:

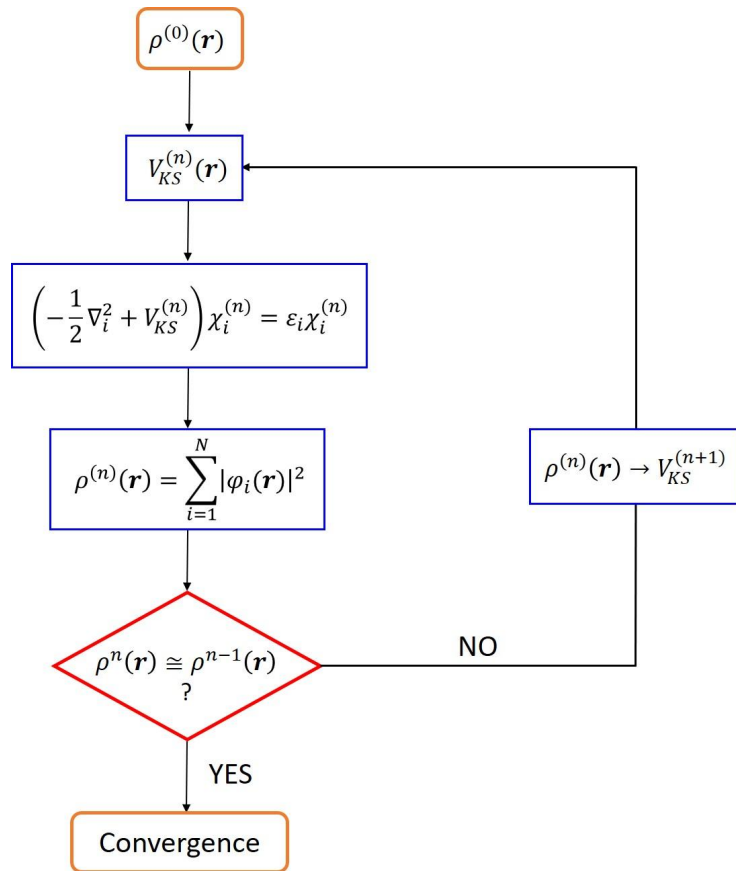
$$V_{KS}(\mathbf{r}) = \int \frac{\rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta E_{XC}}{\delta \rho} + V_{ext}(\mathbf{r}) \quad h_{KS}\chi_i = \varepsilon_i\chi_i$$


The exchange-correlation functional is unknown, but small with respect to the other terms, and must be approximated

Note: V_{KS} depends on the density, which in turn depends on the spatial orbitals...

We need to adopt a **self consistent field** (SCF) procedure to solve the KS equations!

Self consistent field



- We provide an initial density (usually the sum of atomic densities)
- We calculate the KS potential and solve the mono-electronic equations
- The resulting molecular orbitals are used to compute a new electron density
- The new density is compared with that calculated in the previous iteration
- Convergence is reached when the difference between the two density is below a certain threshold

An SCF calculation will be the first exercise of the afternoon!

The exchange-correlation functional

The XC energy functional is the unknown term that we need to approximate

$$E_{xc}[\rho] = T[\rho] - T_0[\rho] + V_{ee}[\rho] - J[\rho]$$

Exchange term: it comes from the antisymmetry of the wavefunction (Pauli principle) which forbids two same-spin electrons from being at the same point in space.

Correlation term: it refers to the additional way electrons avoid each other because of their Coulomb repulsion (on top of what the classical J term already captures).

- Several approximations have been presented over the years
- Different approximations lead to different accuracy but also computational cost

The exchange-correlation functional

- **Local Density Approximation (LDA)**: it describes the electrons with the uniform gas model. Despite being a simple model, it works well in many cases, in particular for geometry optimizations.
- **Generalized Gradient Approximation (GGA)**: it accounts for the non-homogeneity of the electron density by considering its gradient. It improves the calculation of energy-related properties but performs poorly on materials with localized electrons.
- **Meta-GGA**: higher-order derivatives of the electron density for its approximation in the functional energy.
- **Hybrid functionals**: include a fraction of exact (Hartree-Fock) exchange mixed with GGA exchange. It is more accurate than GGA but also more expensive.
- **Range-separated (RS) functionals**: split the electron-electron interaction into short- and long-range components, using exact exchange in one range and DFT exchange in the other. RS functionals describe well non-covalent systems.

Basis set

We discussed the density, which is built from the orbitals:
but how are the orbitals treated in practice?

Orbitals of the KS equations are obtained with the **Linear Combination of Atomic Orbitals** (LCAO) formulation → the orbitals are expressed as linear combinations of basis functions:

$$\varphi_i(\mathbf{r}) = \sum_{k=1}^{Basis} v_k(\mathbf{r}) C_{ki}$$

The problem of obtaining the KS eigenfunctions and eigenvalues is reduced to a generalized diagonalization one:

$$H_{KS}C = SCE$$

Overlap matrix:

$$S_{ij} = \langle v_i | v_j \rangle$$

Practical calculation

$$H_{KS}C = SCE$$

Input:

$$H_{KSij} = \langle v_i | H_{KS} | v_j \rangle$$

$$S_{ij} = \langle v_i | v_j \rangle$$

Output:

$$C_{ij}$$

$$E_{ij} = \varepsilon_i \delta_{ij}$$

Different codes use different approaches to solve this problem:

- Amsterdam Modelling Suite (AMS): numerical construction of the H matrix + Slater basis sets
- Gaussian: analytical construction of the H matrix + Gaussian basis set
- **CP2K**: Gaussians for the orbitals + plane waves for the density and Hartree potential